# Laboratory 02

Ventilation as a bilinear dynamical system

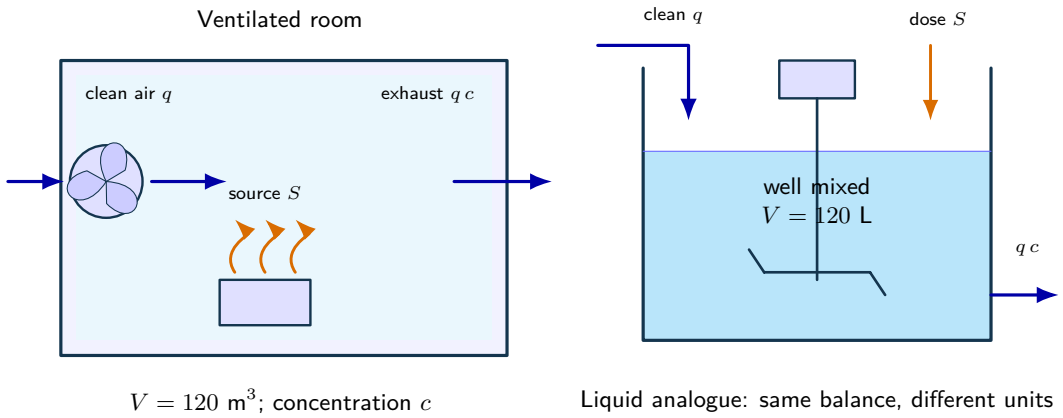
**Exercise sheet – five integrated problems**

**Evidence boundary.** Original hypothetical scenarios. Derive, calculate and justify decisions; every numerical record is synthetic. Figures are conceptual and not to scale. **Shared nominal data.** Nominal room: $V = 120$ m$^3$, source $S = 360$ ppm m$^3$/min and ventilation $q_0 = 3$ m$^3$/min, with clean inlet air and perfect mixing. Excess concentration $c$ is in ppm. Problem 5 introduces a separate liquid analogue with explicitly different units.

**Study route.** On a first modeling pass, solve Problems 1–3 and 5(b). Return to Problem 4 and 5(a,c) after Laboratory 05 introduces PI feedback; revisit actuator limits after Laboratory 09.

## 1. From a ventilated room to a local input–output model

A room has a uniform excess tracer concentration. Fresh air removes tracer, and a separate source generates it. Distinguish airflow from tracer flow in the physical illustration.




A ventilated room and a well-mixed liquid analogue share an inventory balance. The room uses m$^3$ and the liquid case uses L; do not mix their concentration or flow units. Equal inlet and outlet volume flows keep each inventory constant.

**(a)** A well-mixed volume has $V = 120$ m$^3$, source $S = 360$ ppm m$^3$/min and ventilation $q$ in m$^3$/min. Derive the balance for excess concentration $c$ and the equilibrium and time constant at $q_0 = 3$.

**(b)** Linearize $V\dot{c} = S - qc$ about $(S_0, q_0, c_0) = (360, 3, 120)$. Retain source and ventilation deviations as inputs. Give both transfer functions and the term neglected by linearization.

**(c)** For nonzero $\delta c$, compare the discarded bilinear term with the retained term $q_0\delta c$. Give a useful local smallness condition and explain why it is not a complete error bound for a long transient.

**Submission.** Show the model, intermediate reasoning, units and a conclusion addressing the stated requirements.

## 2. Testing a linearization with progressively larger airflow steps

Keep the source fixed and increase airflow from its nominal value. Compare exact equilibria, local predictions and the time needed to cross a concentration threshold.

**(a)** Starting from 120 ppm, increase flow to 3.3 $m^3$/min while retaining the source. Derive the exact trajectory and the local-model prediction. Compare their equilibria and time constants.

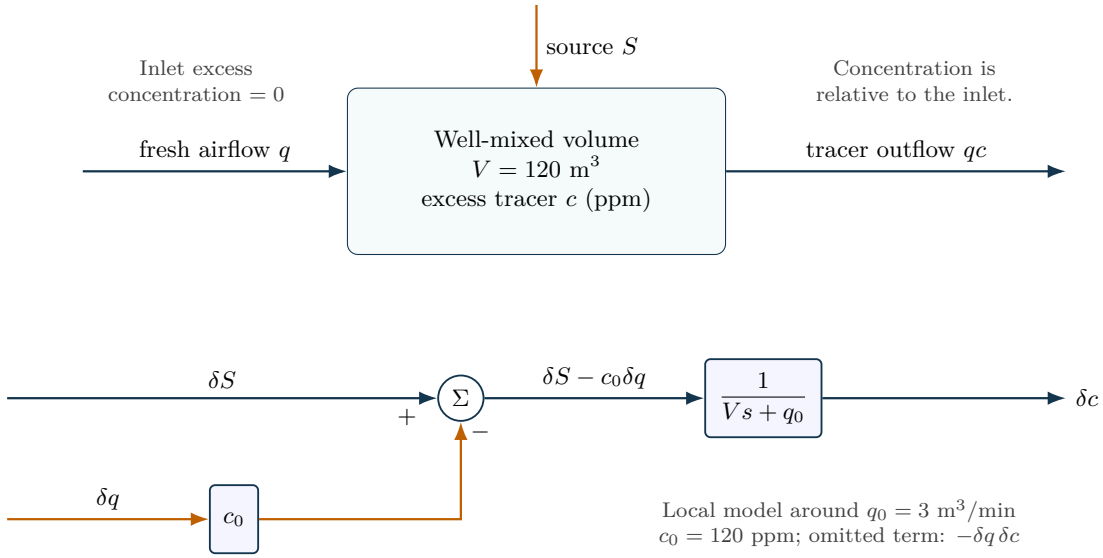
**(b)** Repeat the flow change with $q = 6$. Find both equilibrium predictions and the exact time to reach 75 ppm. Explain whether the zero equilibrium predicted by the local model is physically attainable with a positive source.

**(c)** Let $q = q_0(1 + \varepsilon)$ with $\varepsilon > -1$. Derive exact and linearized equilibrium concentration, their absolute difference and their relative difference measured against the exact concentration. Evaluate the last quantity for $\varepsilon = 0.1$ and 1.

**Submission.** Show the model, intermediate reasoning, units and a conclusion addressing the stated requirements.

## 3. A feedforward specification under source uncertainty

The ventilation actuator has finite capacity. Design decisions must account both for the largest disturbance and for uncertainty in its magnitude.




The physical balance and its local two-input signal model. Airflow enters with a negative sign because increased ventilation removes excess tracer.

**(a)** The target excess concentration is 100 ppm and the source rises from 360 to 480 ppm m$^3$/min. Find the required feedforward flows. If flow is limited to 4 m$^3$/min, find the final concentration after the source change.

**(b)** At fixed flow $q = 3.6$, the source is only known to lie in $[324, 396]$. Find the equilibrium interval and the flow needed to guarantee an equilibrium no larger than 110 ppm under that source uncertainty.

**(c)** Now use the uncertain source interval $[324, 396]$ ppm m$^3$/min, require $c_\infty \leq 100$ ppm and retain maximum flow 4 m$^3$/min. Find the smallest constant flow that guarantees the requirement and the corresponding concentration interval.

**Submission.** Show the model, intermediate reasoning, units and a conclusion addressing the stated requirements.

## 4. Choosing the feedback sign and checking the resulting operating point

The controller increases fresh airflow when excess concentration rises above its target. Use the nonlinear balance to check a controller designed from the local model.

**(a)** Use $q = q_0 + k(c - c_{ref})$ near $c_{ref} = c_0 = 120$. Derive the local closed-loop pole and choose $k$ for a 20 min time constant. Specify the units of $k$.

**(b)** With the proportional law of Problem 4(a), a constant source increment of 60 ppm $m^3$/min occurs. Find the local steady concentration and flow deviations. Then find the exact positive equilibrium of the nonlinear balance and compare.

**(c)** For the 60-unit source increment, calculate the exact equilibrium airflow with the specified proportional controller and check the limit $q \leq 4$. Explain why neither more proportional gain nor integral action can maintain 100 ppm when $S = 480$ and this limit applies.

**Submission.** Show the model, intermediate reasoning, units and a conclusion addressing the stated requirements.

## 5. The same balance in a stirred dilution vessel

Use the liquid analogue shown beside the room: an ideally mixed 120 L vessel receives clean water at $q$ L/min and a negligible-volume solute dose $S = 360$ mg/min. Equal inlet and outlet flows maintain volume. Here $c_{ref} = 120$ mg/L; the gains in part (a) use consistent L, mg and min units.

**(a)** Let $e = c - c_{ref}$, $q = q_0 + k_p e + z$, and $\dot{z} = k_i e$. Derive the linear characteristic polynomial. For $k_p = 0.025$ and $k_i = 0.0004$, calculate natural frequency and damping ratio.

**(b)** Assume constant $S \geq 0$, $q > 0$ and $c(0) \geq 0$. Prove that the exact concentration stays nonnegative and tends to $S/q$. Give two reasons a local model or this mixing model can be misleading in practice.

**(c)** Map the room model to this vessel, state the new concentration units, and evaluate the feasibility of a 100 mg/L target when $S = 480$ mg/min and $q \leq 4$ L/min. Explain the consequence for the proposed integral controller.

**Submission.** Show the model, intermediate reasoning, units and a conclusion addressing the stated requirements.